DUE: FRIDAY, SEPTEMBER 18, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN BEFORE 5:00 PM ON THE DUE DATE. PER THE SYLLABUS, LATE HOMEWORK IS NOT ACCEPTED.

Unless specified otherwise, for all questions in this HW, assume temperature = 300K (kT = 0.026 eV) and $n_i = 1.5 \times 10^{10}$ cm^{-3}.

1. CARRIER CONCENTRATION

(A). Show that the electron and hole distribution functions:

$$n_0 = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) \tag{1.1}$$

$$p_0 = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right) \tag{1.2}$$

are equivalent to

$$n_0 = n_i \exp\left(\frac{E_F - E_i}{k_B T}\right) \tag{1.3}$$

$$p_0 = n_i \exp\left(\frac{E_i - E_F}{k_B T}\right) \tag{1.4}$$

where E_i is the intrinsic energy level.

(B). Derive that

$$n_0 p_0 = n_i^2 \tag{1.5}$$

When is this equation valid?

2. TEMPERATURE EFFECTS

Consider a piece of silicon doped with acceptors $N_A = 1.0 \times 10^{17}\ \text{cm}^{-3}$. Assume complete ionization. At $T = 300$ K, the intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$, and at $T = 500$ K, $n_i = 8.4 \times 10^{13}\ \text{cm}^{-3}$. Use $k_B = 8.617 \times 10^{-5}$ eV/K.

(A). At room temperature, compare the mobilities of two Si samples doped with $N_D = 10^{15}\ \text{cm}^{-3}$ and $N_D = 10^{19}\ \text{cm}^{-3}$. Which one has the higher mobility, and why?

(B). The Fermi level relative to the intrinsic level is

$$E_F - E_i = -k_B T \ln\left(\frac{N_A}{n_i}\right) \tag{2.1}$$

Calculate $E_F - E_i$ at $T = 300$ K and $T = 500$ K. Comment on how the Fermi level shifts with temperature.

(C). Using charge neutrality together with eq. (1.5), calculate the equilibrium electron concentration n_0 in this sample at $T = 300$ K and $T = 500$ K. Comment on whether the semiconductor is extrinsic or intrinsic at each temperature.

3. BAND DIAGRAM

A piece of Si uniformly doped with acceptors at a concentration of $N_A = 5 \times 10^{13}\ \text{cm}^{-3}$ is at equilibrium at $T = 300$ K. Assume complete ionization of dopants. The intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$.

(A). Write the charge neutrality equation for this semiconductor and use it to determine the equilibrium hole and electron concentrations.

(B). Where is the Fermi level E_F relative to the intrinsic level E_i? Calculate $E_i - E_F$.

(C). Draw the equilibrium energy band diagram for this silicon. The horizontal axis is position, and the vertical axis is energy. Clearly label E_F, E_i, E_C, and E_V and indicate their relative positions.

4. RESISTANCE UNDER DOPING

A Si bar has a length of 10 mm and a cross-sectional area of 1 mm^2. At $T = 300$ K, the intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}\ cm^{-3}$. Read the mobility values from the provided figure. Calculate its resistance in the following cases:

(A). Without any doping.

(B). Doped with $5 \times 10^{17}\ cm^{-3}$ phosphorus.

(C). Doped with $5 \times 10^{17}\ cm^{-3}$ phosphorus and $2 \times 10^{16}\ cm^{-3}$ boron.

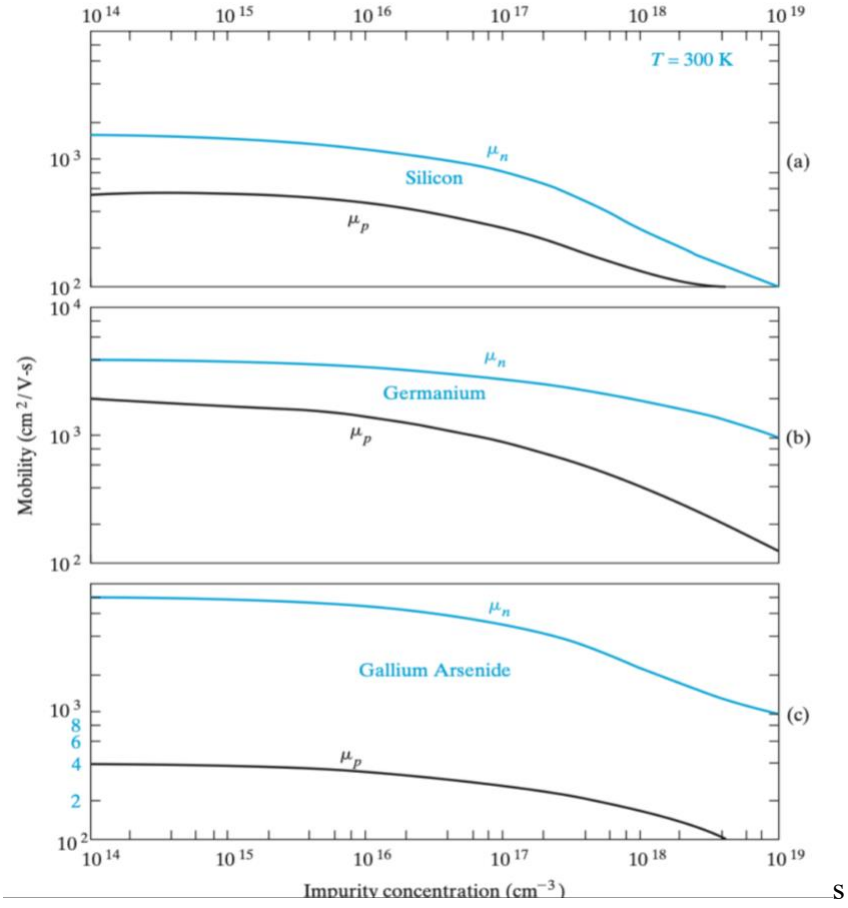


Figure 3-23: Variation of mobility with total doping impurity concentration $(N_a + N_d)$ for Ge, Si, and GaAs at 300K.